Summary Table: Privacy-Preserving Techniques Cited in Khalid et al. (2023)

This table compares some important studies cited in Khalid et al. (2023). I selected these references because they represent the main privacy-preserving techniques discussed in the paper, including encryption, differential privacy, federated learning, blockchain, and hybrid methods.

Cited Reference	Type	Technique	Healthcare Use	Strength	Limitation / Gap
Chowdhury et al. [68]	Cryptographic	DeCrypt / Encryption	Secure healthcare data sharing	Faster than 3DES and gives better protection	May be difficult to use in practice
Sarkar et al. [69]	Cryptographic	Homomorphic Encryption	Private genome analysis	Allows analysis without exposing raw data	Can be slow and expensive
Paul et al. [70]	Cryptographic	Homomorphic Encryption	ICU data model training	Protects patient data during training	Needs more testing in real hospital settings
Akgun et al. [82]	Cryptographic	SMPC	Private genomic database search	Allows private collaboration between parties	Communication cost can be high
Li et al. [84]	Cryptographic	SMPC	Secure medical computation	Raw data does not need to be shared directly	Depends on trust between servers
Jiang et al. [75]	Non-cryptographic	Differential Privacy	Data publishing / model output	Protects individual patient records	Too much noise may reduce accuracy
Khan et al. [76]	Federated	Federated Learning	Multi-hospital AI training	Data can stay inside each hospital	Model updates may still leak information
Nakamoto [77]	Decentralized	Blockchain	EHR sharing and audit trail	Helps with transparency and access control	Can be hard to scale
Ngan et al. [116]	Hybrid	FL + DP + Blockchain	Private medical data sharing	Combines multiple privacy protections	More complex to implement
Alzubi et al. [115]	Hybrid	DL + Blockchain + FL	EMR privacy and access control	Supports AI use while protecting records	May be difficult to set up

From this comparison, I found that each privacy-preserving technique has a different role in healthcare AI. Encryption-based methods provide strong protection, but they can be slow and complex. Differential Privacy protects individual patient records, but too much noise may reduce accuracy. Federated Learning is useful because hospitals can train AI models without sharing raw patient data, but it is still not completely safe from attacks. Overall, I think hybrid methods are more practical because one technique alone may not solve all privacy and security problems.